

Investor Sentiment and the Pricing of Behavioral Beta in U.S. Equity Markets

Sivagurunathan Nathan, Srija Balaraju, Pavithraa Sivakumar

Grenoble Ecole de Management

March 03, 2026

Abstract

This study examines whether investor sentiment and sentiment-related risk exposure help explain the returns of major U.S. sector exchange-traded funds between 2000 and 2023. The analysis draws on monthly return data for five widely traded sector ETFs: Energy, Financials, Consumer Staples, Technology, and Utilities (XLE, XLF, XLP, XLK, XLU). It also incorporates a Baker-Wurgler style sentiment index and evaluates sentiment within several empirical frameworks, including the Capital Asset Pricing Model, behavioral beta regressions, predictive return models, high-low behavioral beta spread tests, regime interaction specifications, multifactor robustness checks, behavioral-beta-only regressions and pooled panel regressions. An orthogonalized sentiment measure is used to separate behavioral influences from market related movements. Across eight hypotheses, the results indicate that market beta remains an important determinant of sector fund returns, while behavioral beta appears modest, varies across sectors, and does not generate a consistent return premium. Sentiment shows limited ability to forecast future sector ETF returns, and its influence is largely absorbed by standard risk factors in a multifactor setting. Taken together, the findings suggest that sentiment-related risk is not systematically priced at the sector ETF level and that traditional asset pricing models continue to describe return patterns in this context. Future research may extend this framework to individual stocks, alternative sentiment indicators, or higher frequency data to identify conditions under which sentiment becomes more economically relevant.

1. Introduction

Traditional asset pricing theory posits that expected returns compensate investors for bearing systematic risk. The Capital Asset Pricing Model (CAPM) formalizes this relationship by linking expected returns to market beta, which measures an asset's sensitivity to aggregate market movements (Sharpe, 1964). This framework assumes rational investors, frictionless markets, and efficient information incorporation.

Empirical evidence, however, challenges these assumptions. Behavioral finance studies show that investors are influenced by overconfidence, optimism, pessimism, and herding. These biases can distort risk perceptions, generate temporary mispricing, and weaken the traditional risk-return relationship.

Investor sentiment, referring to investor beliefs about future returns not fully explained by fundamentals, may distort pricing. De Long et al. (1990) demonstrate that when arbitrage is limited or risky, noise traders can sustain mispricing and influence equilibrium returns. In this context, sentiment can affect aggregate risk pricing rather than merely introduce transitory noise.

Empirical studies suggest that sentiment also impacts the pricing of systematic risk. Verma and Soydemir (2009) show that irrational sentiment compresses the market price of risk. Antoniou et al. (2016), Shen et al. (2017), and Wang (2020) document that the beta-return relation varies across sentiment regimes, with high-beta stocks underperforming during optimistic periods. These findings imply that behavioral beta, defined as an asset's sensitivity to sentiment-driven fluctuations after controlling for market movements, may represent an additional dimension of risk.

While prior research largely focuses on individual stocks, sector exchange-traded funds (ETFs) remain underexplored. By pooling individual stocks, sector ETFs maintain systematic exposures associated with economic cycles and investor preferences. The substantial institutional ownership of sector ETFs could mitigate or alter sentiment-driven effects relative to single-stock behavior. Differences in growth potential, cyclicalities, and speculative appeal suggest that high-beta sectors such as technology or energy may exhibit stronger behavioral betas, while defensive sectors such as utilities may be less sensitive.

This motivates examination of whether behavioral beta is systematically priced, whether it is independent of market beta, and whether it retains significance in multifactor frameworks. Using

approximately twenty-three years of monthly data for five major U.S. sector ETFs, namely Energy, Financials, Consumer Staples, Technology, and Utilities, this study addresses six research questions:

1. Does the traditional market beta-return relation hold for sector ETFs?
2. Do sector ETFs exhibit statistically significant behavioral betas?
3. Does investor sentiment predict future sector ETF returns?
4. Is sentiment-related risk associated with a return premium?
5. Does behavioral beta remain significant within a multifactor framework?
6. Is there a systematic relationship between market beta and behavioral beta across sectors?

To address these questions, eight empirical hypotheses are developed and evaluated through a complementary testing framework. The first examines whether the conventional CAPM beta-return relationship holds at the sector level. The second introduces behavioral beta within a market and sentiment specification to assess whether sentiment exposure carries a premium. Additional tests consider predictive regressions of future returns, high-low behavioral beta spread portfolios, and regime-dependent interaction models designed to capture state variation in risk pricing. Robustness is further evaluated using the Fama-French five-factor model, pooled panel regressions across sectors, and a specification in which behavioral beta alone is used to explain returns. Although monthly data provide a long-horizon perspective on pricing relations, higher-frequency measures may capture more immediate sentiment dynamics, which remain beyond the scope of the present analysis.

The remainder of the paper is organized as follows. Section 2 reviews the literature on investor sentiment, behavioral beta, and sector-level asset pricing. Section 3 describes the data and construction of key variables. Section 4 presents the econometric methodology and eight empirical tests. Section 5 reports the main results and robustness checks. Section 6 discusses the implications of our findings and outlines limitations and avenues for further research. Section 7 concludes.

2. Literature Review

2.1 Investor Sentiment and the Market Price of Risk

Models of sentiment-based pricing consider markets where rational arbitrageurs operate alongside noise traders. De Long et al. (1990) demonstrate that when arbitrage is costly or risky, mispricing induced by sentiment can persist and affect equilibrium returns, making sentiment risk systematic.

Verma and Soydemir (2009) further show that irrational optimism reduces the market price of risk, whereas rational sentiment aligned with fundamentals has limited effect. Distinguishing information-driven expectations from psychologically driven optimism is critical for understanding pricing implications. Baker and Wurgler (2006) show that comprehensive sentiment measures correlate with overall returns and factor premiums, reinforcing the view that sentiment can act as a priced source of systematic risk rather than purely diversifiable noise.

2.2 Sentiment and the Beta-Return Relation

CAPM predicts a positive beta-return relation, yet Fama-French (1992) find it weak or flat in cross-sectional data. Behavioral explanations suggest that speculative demand drives high-beta stocks to become overpriced during optimistic periods (Antoniou et al., 2016). Shen et al. (2017) report that macroeconomic risk premia also vary across sentiment regimes.

Wang (2020) finds that institutional sentiment can reverse the slope of the Security Market Line: beta is positively priced in pessimistic periods but negatively priced in optimistic ones. These findings indicate that sensitivity to sentiment, or behavioral beta, may provide incremental explanatory power beyond traditional beta. Taken together, this evidence implies that tests of the standard CAPM beta-return relation (H1) may yield different results across sentiment conditions, and that a behavioral beta capturing sensitivity to sentiment innovations (H2) may provide additional pricing information beyond traditional market beta.

2.3 Institutional Trading and Asset Pricing

Frijns et al. (2018) show that beta-return relations are positive on days with high institutional trading and negative when activity is low, suggesting that market composition and trading intensity influence risk pricing. For sector ETFs, which are held by both institutional and retail investors, these dynamics may create variation in behavioral beta across sectors. Given the high institutional ownership of many sector ETFs, these results suggest that cross-sector differences in behavioral beta and any aggregate sentiment effect at the ETF level (H2, H7) are likely to depend on how institutional investors trade around sentiment shocks.

2.4 Sentiment and the Mean-Variance Tradeoff

Sentiment also affects the risk-return tradeoff. Yu and Yuan (2011) find that high sentiment weakens the positive mean-variance relation. Wang and Duxbury (2021) extend this internationally, showing stronger distortions in markets with weaker investor protection or greater susceptibility to behavioral bias. These findings imply that sentiment may shape the fundamental relation between risk and return in diversified portfolios. From an empirical perspective, this raises the question of whether sentiment retains explanatory power once standard risk factors are controlled for and whether its effect is state-dependent, motivating our regime-interaction tests (H5) and Fama-French five-factor robustness tests (H6).

2.5 Contribution of the Present Study

While prior studies show that sentiment influences the market price of risk, beta premia, and aggregate risk-return dynamics, sector ETFs remain relatively unexamined. This study contributes by estimating behavioral beta at the sector level and testing whether sentiment exposure is priced within both CAPM and multifactor frameworks. It also examines whether market beta and behavioral beta are systematically related, shedding light on whether sentiment amplifies existing risk exposures or represents an independent dimension. Methodologically, we integrate eight complementary tests (H1-H8) including CAPM, behavioral beta regressions, predictive regressions, high-low spread portfolios, regime interactions, Fama-French five-factor models, pooled panel regressions, and behavioral-beta-only specifications within a single coherent framework. By focusing on actively traded sector ETFs rather than individual stocks, the paper assesses whether sentiment-driven risk is still priced at an aggregate, institutionally dominated level of the U.S. equity market.

3. Data and Variables

3.1 Sample and Data Sources

The dataset consists of approximately 288 monthly observations spanning January 2000 to December 2023. The sample includes five U.S. sector ETFs: XLE (Energy), XLF (Financials), XLP (Consumer Staples), XLK (Technology), XLU (Utilities). Market returns are measured using the S&P 500 (SPX) total return index, which includes dividends. The risk-free rate is proxied by the one-month U.S. Treasury bill rate. Investor sentiment is captured using a Baker-Wurgler type

orthogonalized sentiment index. In addition, the Fama-French five factors (SMB, HML, RMW, CMA and the market factor) are obtained from Kenneth French's data library.

These data sources provide a consistent foundation for analyzing sector ETF behavior in relation to market and sentiment factors over the study period.

3.2 Construction of Returns and Excess Returns

To align with asset pricing theory, all analyses are conducted using excess returns. The risk-free rate was converted from percentage to decimal form and the data were arranged chronologically in time-series format. Excess returns were computed as follows:

$$R_{i,t}^{excess} = R_{i,t} - R_{f,t}$$

$$R_{m,t}^{excess} = R_{m,t} - R_{f,t}$$

Where $(R_{i,t})$ denotes ETF returns, $(R_{m,t})$ denotes market returns (SPX) and $(R_{f,t})$ represents the one-month Treasury bill rate. Extreme observations were not trimmed to preserve the integrity of the full dataset.

3.3 Sentiment Index and Orthogonalization

Investor sentiment is measured using an orthogonalized Baker-Wurgler type sentiment index. The original index is constructed from market-based proxies, including IPO activity, trading volume, equity issuance and closed end fund discounts. The sentiment measure is orthogonalized with respect to macroeconomic and market-wide control variables, removing components related to fundamentals.

The orthogonalization can be represented as:

$$Sentiment_t = \alpha + \gamma Z_t + u_t$$

Where (Z_t) denotes macroeconomic and market controls. The residual term (u_t) constitutes the orthogonalized sentiment measure (SENT_ORTH), capturing the behavioral variation that is independent of macroeconomic factors. This ensures that SENT_ORTH reflects sentiment-driven deviations rather than fundamentals.

3.4 Descriptive statistics

Table 1: Descriptive statistics of returns and sentiment

Panel A presents summary statistics for ETF excess returns and market excess return over the period January 2000 to December 2023. Panel B reports summary statistics for the orthogonalized sentiment index.

Table 1 Panel A: ETF and Market Excess Returns

	Mean	Std. Dev.	Min	Max
XLE_excess	0.007495	0.074908	-0.359274	0.335700
XLF_excess	0.004861	0.062877	-0.250962	0.229950
XLP_excess	0.005264	0.036257	-0.121637	0.097027
XLK_excess	0.006606	0.066558	-0.254462	0.246276
XLU_excess	0.005375	0.044596	-0.140271	0.117827
SPX_excess	0.003978	0.044643	-0.170925	0.125644
SENT_ORTH	0.156925	0.690050	-0.967515	3.019232

Table 1 Panel B: Sentiment

	SPX_excess	SENT_ORTH
XLE_excess	0.631569	0.013370
XLF_excess	0.838048	-0.079752
XLP_excess	0.674246	-0.068673
XLK_excess	0.861748	-0.197018
XLU_excess	0.501952	-0.065922
SPX_excess	1.000000	-0.157318
SENT_ORTH	-0.157318	1.000000

Standard Deviation of Sentiment: 0.6900498484603167

Note: This table reports monthly summary statistics for the period January 2000 to December 2023.

These descriptive statistics indicate that volatility is higher for cyclical sectors such as XLE and XLK, whereas defensive sectors like XLU and XLP show lower variability.

Table 2: Correlation matrix

Table 2 Correlation Matrix (ETF returns)

	XLE_excess	XLF_excess	XLP_excess	XLK_excess	XLU_excess
XLE_excess	1.000000	0.567505	0.418785	0.418348	0.366098
XLF_excess	0.567505	1.000000	0.572958	0.575457	0.389807
XLP_excess	0.418785	0.572958	1.000000	0.422319	0.628987
XLK_excess	0.418348	0.575457	0.422319	1.000000	0.310646
XLU_excess	0.366098	0.389807	0.628987	0.310646	1.000000
SPX_excess	0.631569	0.838048	0.674246	0.861748	0.501952
SENT_ORTH	0.013370	-0.079752	-0.068673	-0.197018	-0.065922

Note: This table reports monthly correlation coefficients for the period January 2000 to December 2023.

Table 2 presents pairwise correlations among ETF excess returns, the market excess returns, and the orthogonalized sentiment index.

ETFs are positively correlated with the market particularly XLK(0.86) and XLF(0.84), indicating high systematic exposure. Cross-sector correlations are moderate to high, reflecting common market influences. The orthogonalized sentiment index exhibits weak correlations with ETF excess returns, ranging from -0.20(XLK) to 0.01(XLE), suggesting limited linear association at the monthly horizon. Overall, these patterns confirm that ETF returns are primarily driven by market factors, with sentiment capturing an orthogonal behavioral component that is weakly correlated with returns.

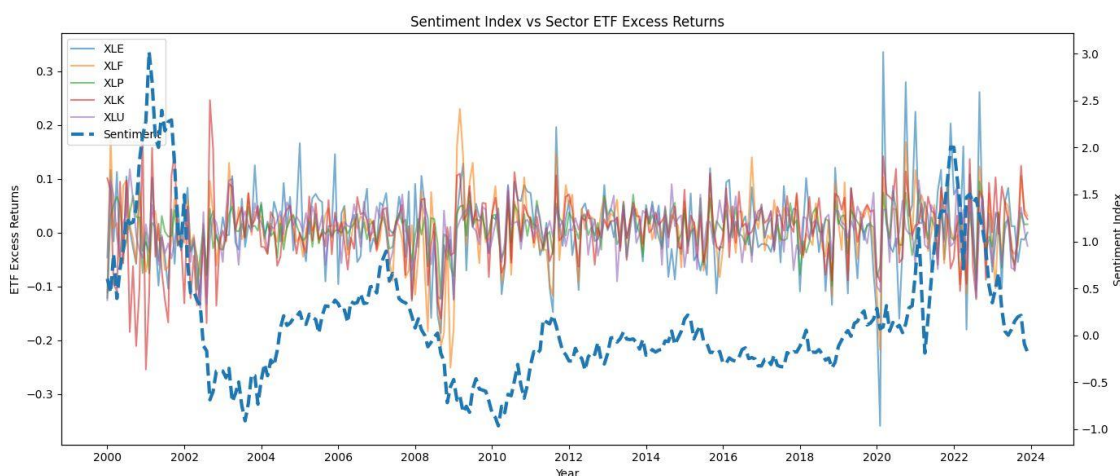


Figure 1. Sentiment index (SENT_ORTH) and sector ETF excess returns over time.

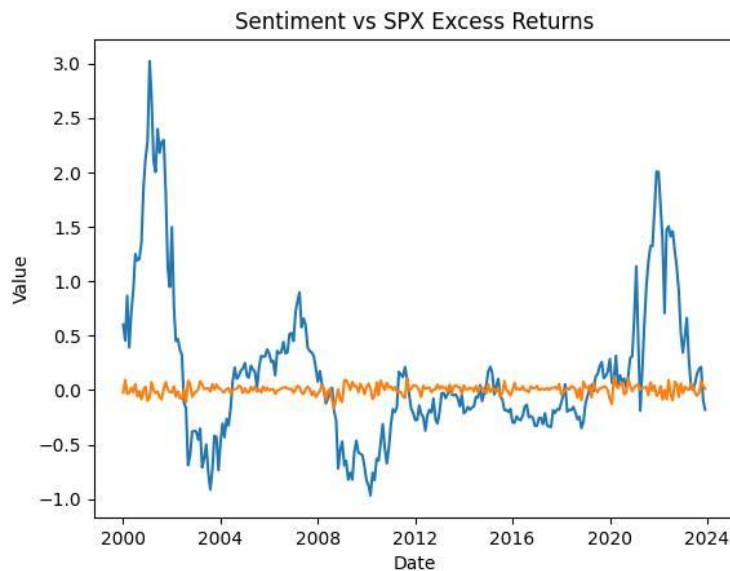


Figure 2. Orthogonalized sentiment index vs S&P 500 excess returns.

4. Methodology

This study examines whether investor sentiment affects how behavioral beta is priced in the U.S. stock market, using empirical research. To do this, we used eight empirical tests. These tests are divided into two main categories. The first set of pricing tests, hypotheses 1-4 test the connection between sentiment and returns, as well as whether behavioral beta is included in the pricing model. The second set includes robustness and extended standards. Hypotheses 5-8 assess whether the impact of sentiment remains stable, persists in pooled market-level regressions, and is still significant after adjusting for known asset-pricing determinants.

The empirical analysis uses monthly excess returns from January 2000 to December 2023 for five U.S. sector ETFs: Energy (XLE), Financials (XLF), Consumer Staples (XLP), Technology (XLK), and Utilities (XLU). These excess returns are calculated by subtracting the one-month U.S. Treasury bill rate. Market returns are represented by the S&P 500 total return index. Additionally, investor sentiment is measured using a composite sentiment index. This index is adjusted to remove macroeconomic factors, ensuring that the sentiment variable reflects behavioral influences rather than fundamental risk.

4.1 CAPM Baseline

The analysis begins with estimation of the Capital Asset Pricing Model (CAPM) for each of the ETFs. The baseline model is specified as follows:

$$R_{i,t}^{excess} = \alpha_i + \beta_{m,i} R_{m,t}^{excess} + \varepsilon_{i,t}$$

where $R_{i,t}^{excess}$ denotes the excess return of ETF i in month t , $R_{m,t}^{excess}$ and represents the market excess return.

This model tests the validity of H1, which states that market beta is a significant determinant of ETF excess return. This model is based on the CAPM, which is used as a benchmark model. By employing the CAPM, the research is able to establish the validity of the model before incorporating behavioral factors. It validates the CAPM before examining the effect of sentiments.

4.2 Behavioral Beta Regression

To examine whether investor sentiment is priced in ETF returns, the CAPM is extended as follows:

$$R_{i,t}^{excess} = \alpha_i + \beta_{m,i} R_{m,t}^{excess} + \beta_{s,i} Sentiment_t + \varepsilon_{i,t}$$

In this specification, $\beta_{s,i}$ represents the behavioral beta, capturing the sensitivity of ETF returns to changes in investor sentiment.

H2 proposes that beta of behavioral risk is reflected in the market model. A statistically significant coefficient for the sentiment variable indicates that sentiment risk explains a portion of the return beyond market risk. This is a direct test of whether behavioral risk provides additional explanatory power beyond CAPM.

4.3 Predictive Regression

The predictive properties of investor sentiment are evaluated using the following regression:

$$R_{i,t+1}^{excess} = \alpha_i + \gamma_i Sentiment_t + \varepsilon_{i,t+1}$$

This model examines whether current sentiment predicts next month's ETF excess returns.

H3 states that investor sentiment contains predictive information about future returns. The significance of the coefficient implies that the behavioral effects are not only contemporaneous but have forecasting ability. This test is particularly relevant for understanding whether sentiment mispricing persists and corrects gradually over time.

4.4 High-Low Behavioral Beta Spread Test

The economic significance of behavioral beta is further evaluated using exchange-traded funds (ETFs), which are sorted based on their estimated β_s . They are then sorted into high and low behavioral beta portfolios.

The return spread is calculated as:

$$Spread_t = R_{High,t}^{excess} - R_{Low,t}^{excess}$$

The average spread is tested using:

$$Spread_t = \alpha + \varepsilon_t$$

H4 states that high behavioral beta ETFs earn different average returns than low behavioral beta ETFs. A statistically significant spread would indicate that sentiment-sensitive ETFs exhibit abnormal performance compared to those less exposed to sentiment. This portfolio-based approach is a complement to the regression-based results and shows the economic significance of behavioral pricing effects.

4.5 Regime / Interaction Model

To determine whether sentiment effects vary across market conditions, a high-volatility dummy variable is constructed. The dummy variable is equal to one if the volatility exceeds the historical 75th percentile, and zero otherwise.

The interaction model is estimated as:

$$R_{i,t}^{excess} = \alpha_i + \beta_{m,i}R_{m,t}^{excess} + \beta_{s,i}Sentiment_t + \beta_{int,i}(Sentiment_t \times HighVol_t) + \varepsilon_{i,t}$$

H5 examines whether the influence of sentiment intensifies during high-volatility periods. A statistically significant interaction term would suggest that behavioral effects become more pronounced when market uncertainty rises.

4.6 Fama-French Five-Factor Robustness

To verify that sentiment effects are not driven by conventional risk factors, the Fama-French five-factor model is estimated using the following equation:

$$R_{i,t}^{excess} = \alpha_i + \beta_m R_{m,t}^{excess} + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \beta_{RMW} RMW_t + \beta_{CMA} CMA_t + \beta_s Sentiment_t + \varepsilon_{i,t}$$

This specification controls for size, value, profitability, and investment factors.

H6 states that the sentiment effect is significant after controlling for the Fama-French five factors. If the coefficient on the sentiment effect is statistically significant, then the behavioral beta picks up a unique source of return variation not explained by conventional asset pricing factors.

4.7 Panel Regression and Aggregate Effect

In order to test whether the effect of sentiment is generalized across sectors, a panel regression of the following type is estimated:

$$R_{i,t}^{excess} = \alpha + \beta_m R_{m,t}^{excess} + \beta_s \text{Sentiment}_t + \varepsilon_{i,t}$$

This model pools data across ETFs and estimates whether sentiment affects returns at the aggregate ETF market level.

H7 proposes that investor sentiment significantly influences ETF excess returns in a pooled framework. A significant pooled sentiment coefficient would imply that the effects of behavioral pricing are not limited to a particular industry; rather, they represent a broader market phenomenon.

4.8 Behavioral Beta Only Regression

To further assess whether investor sentiment independently explains ETF returns, a behavioral beta-only specification is estimated without including the market factor:

$$R_{i,t}^{excess} = \alpha_i + \beta_{s,i} \text{Sentiment}_t + \varepsilon_{i,t}$$

This model removes market excess returns from the regression in order to evaluate whether sentiment captures an independent pricing component or merely proxies for market risk.

H8 states that investor sentiment independently explains excess returns even without controlling for market risk. The statistical significance of the sentiment coefficient determines whether this condition holds.

4.9 Estimation Procedure

All regressions are estimated using Ordinary Least Squares (OLS) with heteroskedasticity robust standard errors using the HC3 method. The analysis is conducted on a monthly basis. Although return persistence and autocorrelation are often low at monthly intervals, these limitations are acknowledged.

Regression outputs report estimated coefficients, t-statistics, R^2 values, and statistical significance levels. The empirical analysis was carried out using Python, which helped ensure both reproducibility and consistency across different specifications.

5. Results

5.1 CAPM Validation

The sector ETF results are primarily driven by systematic market risk, and sectors like utilities and consumer staples exhibit lower sensitivity, while technology and financial sector ETFs show higher market exposure. All estimated market betas are positive and statistically significant, with XLU displaying the lowest beta and XLK and XLF the highest, consistent with their defensive and cyclical profiles. Overall, the results support H1, confirming that the CAPM beta-return relationship applies at the sector ETF level.

Table 3. CAPM regression results for sector ETFs.

ETF	Market Beta	p value	R squared
0 XLE	1.059716	5.302877e-23	0.398879
1 XLF	1.180328	1.893576e-93	0.702324
2 XLP	0.547586	8.885279e-42	0.454607
3 XLK	1.284772	1.009251e-98	0.742609
4 XLU	0.501417	1.410729e-16	0.251956

5.2 Behavioral beta estimation

Investor sentiment affects certain sectors rather than all sectors. The energy sector (XLE) appears more sensitive to sentiment fluctuations, possibly due to its cyclical and commodity-driven nature, and it is the only ETF with a statistically significant behavioral beta at conventional levels. For the remaining ETFs, sentiment coefficients are small and insignificant, and the incremental R^2 gain over CAPM is limited. Evidence for H2 is limited, appearing only in the energy sector, suggesting that behavioral beta is not systematically priced across ETFs.

Table 3. Behavioral beta regression results (market + sentiment).

ETF	Alpha	Market Beta	Behavioral Beta	Sentiment pvalue	R squared
0 XLE	0.001189	1.090227	0.012547	0.034946	0.411909
1 XLF	-0.000646	1.192162	0.004867	0.157598	0.705106
2 XLP	0.002750	0.552486	0.002015	0.522289	0.456041
3 XLK	0.002507	1.269994	-0.006077	0.228401	0.746481
4 XLU	0.003236	0.503519	0.000864	0.801695	0.252130

5.3 Predictive regression

When testing whether sentiment predicts next month's results, we found that most ETFs exhibit insignificant predictive coefficients, and the technology sector (XLK) shows weak significance at the 10% level, but the associated economic magnitude is small and not robust across specifications. Investor sentiment therefore does not provide strong forecasting power for future sector ETF returns at the monthly horizon, and H3 is largely rejected.

Table 5. One month ahead predictive regressions of ETF return on sentiment.

ETF	Predict_Coeff	p_value
0 XLE	0.003148	0.629537
1 XLF	-0.005702	0.275771
2 XLP	-0.003492	0.290200
3 XLK	-0.016139	0.079060
4 XLU	-0.004006	0.299015

5.4 High-low behavioral beta spread

The high-low behavioral beta spread test shows that average return difference between high and low behavioral beta ETFs is economically small and statistically insignificant, with a spread alpha of about 0.0009 and a p-value above 0.80. The results indicate that behavioral beta is not associated with a return premium and that investors do not earn compensation for bearing sentiment risk, so H4 is rejected.

Table 6. High-low behavioral beta spread regression (HC3 standard errors).

OLS Regression Results						
Dep. Variable:	Spread		R-squared:	0.000		
Model:	OLS		Adj. R-squared:	0.000		
Method:	Least Squares		F-statistic:	nan		
Date:	Thu, 05 Mar 2026		Prob (F-statistic):	nan		
Time:	11:46:34		Log-Likelihood:	331.71		
No. Observations:	288		AIC:	-661.4		
Df Residuals:	287		BIC:	-657.8		
	Df Model:			0		
	Covariance Type:			HC3		
coef	std err	z	P> z		[0.025	0.975]
const	0.0009	0.005	0.197	0.844	-0.008	0.010
Omnibus:	13.963		Durbin-Watson:	2.037		
Prob(Omnibus):	0.001		Jarque-Bera (JB):	27.820		
Skew:	0.213		Prob(JB):	9.10e-07		
Kurtosis:	4.462		Cond. No.	1.00		

Notes: [1] Standard Errors are heteroscedasticity robust (HC3)

5.5 Regime / Interaction Model

The regime and interaction analysis shows that market stress does not amplify the impact of sentiment on sector ETF returns. The interaction term between sentiment and the high-volatility indicator is insignificant for all ETFs, suggesting no strong regime dependence, and professional trading appears to limit observable behavioral biases in crisis periods. Thus, the influence of sentiment remains broadly stable across different market conditions, and H5 is rejected.

Table 7. Interaction regressions of ETF return on sentiment and high volatility regimes.

ETF: XLE						
OLS Regression Results						
Dep. Variable:	XLE_excess	R-squared:				0.414
Model:	OLS	Adj. R-squared:				0.407
Method:	Least Squares	F-statistic:				33.75
Date:	Thu, 05 Mar 2026	Prob (F-statistic):				1.10e-18
Time:	11:46:34	Log-Likelihood:				415.06
No. Observations:	288	AIC:				-822.1
Df Residuals:	284	BIC:				-807.5
	Df Model:					3
	Covariance Type:					HC3
coef	std err	z	P> z		[0.025	0.975]
const	0.0012	0.003	0.363	0.717	-0.005	0.008
SPX_excess	1.0977	0.110	9.992	0.000	0.882	1.313
SENT_ORTH	0.0083	0.007	1.249	0.212	-0.005	0.021
Interaction	0.0088	0.012	0.707	0.480	-0.016	0.033
Omnibus:		22.881	Durbin-Watson:			2.120
Prob(Omnibus):		0.000	Jarque-Bera (JB):			55.132
Skew:		0.349	Prob(JB):			1.07e-12
Kurtosis:		5.026	Cond. No.			23.6

Notes: [1] Standard Errors are heteroscedasticity robust (HC3)

ETF: XLF						
OLS Regression Results						
Dep. Variable:	XLF_excess	R-squared:				0.707
Model:	OLS	Adj. R-squared:				0.704
Method:	Least Squares	F-statistic:				147.2
Date:	Thu, 05 Mar 2026	Prob (F-statistic):				1.51e-57
Time:	11:46:34	Log-Likelihood:				565.56
No. Observations:	288	AIC:				-1123.
Df Residuals:	284	BIC:				-1108.
	Df Model:					3
	Covariance Type:					HC3
coef	std err	z	P> z		[0.025	0.975]
const	-0.0006	0.002	-0.301	0.764	-0.005	0.004
SPX_excess	1.1993	0.060	19.943	0.000	1.081	1.317
SENT_ORTH	0.0008	0.004	0.179	0.858	-0.008	0.009
Interaction	0.0084	0.007	1.222	0.222	-0.005	0.022
Omnibus:		24.034	Durbin-Watson:			1.957

Prob(Omnibus):	0.000	Jarque-Bera (JB):	95.191
Skew:	-0.047	Prob(JB):	2.14e-21
Kurtosis:	5.815	Cond. No.	23.6

Notes: [1] Standard Errors are heteroscedasticity robust (HC3)

ETF: XLP

OLS Regression Results						
Dep. Variable:	XLP_excess	R-squared:	0.457			
Model:	OLS	Adj. R-squared:	0.451			
Method:	Least Squares	F-statistic:	66.35			
Date:	Thu, 05 Mar 2026	Prob (F-statistic):	1.52e-32			
Time:	11:46:34	Log-Likelihood:	635.13			
No. Observations:	288	AIC:	-1262.			
Df Residuals:	287	BIC:	-1248.			
	Df Model:		3			
	Covariance Type:		HC3			
coef	std err	z	P> z	[0.025	0.975]	
const	0.0028	0.002	1.794	0.073	-0.000	0.006
SPX_excess	0.5553	0.040	13.988	0.000	0.477	0.633
SENT_ORTH	0.0004	0.005	0.091	0.927	-0.009	0.009
Interaction	0.0033	0.007	0.502	0.616	-0.010	0.016
Omnibus:	12.995	Durbin-Watson:	1.990			
Prob(Omnibus):	0.002	Jarque-Bera (JB):	27.155			
Skew:	-0.153	Prob(JB):	1.27e-06			
Kurtosis:	4.473	Cond. No.	23.6			

Notes: [1] Standard Errors are heteroscedasticity robust (HC3)

ETF: XLK						
OLS Regression Results						
Dep. Variable:	XLK_excess	R-squared:	0.750			
Model:	OLS	Adj. R-squared:	0.747			
Method:	Least Squares	F-statistic:	162.4			
Date:	Thu, 05 Mar 2026	Prob (F-statistic):	2.59e-61			
Time:	11:46:34	Log-Likelihood:	571.78			
No. Observations:	288	AIC:	-1136.			
Df Residuals:	287	BIC:	-1121.			
	Df Model:		3			
	Covariance Type:		HC3			
coef	std err	z	P> z	[0.025	0.975]	
const	0.0025	0.002	1.350	0.177	-0.001	0.006
SPX_excess	1.2607	0.060	21.035	0.000	1.143	1.378
SENT_ORTH	-0.0008	0.007	-0.101	0.919	-0.015	0.014
Interaction	-0.0110	0.010	-1.058	0.290	-0.031	0.009
Omnibus:	40.345	Durbin-Watson:	2.213			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	144.332			
Skew:	0.528	Prob(JB):	4.56e-32			
Kurtosis:	6.304	Cond. No.	23.6			

Notes: [1] Standard Errors are heteroscedasticity robust (HC3)

ETF: XLU						
OLS Regression Results						
Dep. Variable:	XLU_excess			R-squared:	0.252	
Model:	OLS			Adj. R-squared:	0.244	
Method:	Least Squares			F-statistic:	22.73	
Date:	Thu, 05 Mar 2026			Prob (F-statistic):	3.21e-13	
Time:	11:46:34			Log-Likelihood:	529.40	
No. Observations:	288			AIC:	-1051.	
Df Residuals:	284			BIC:	-1036.	
	Df Model:				3	
	Covariance Type:				HC3	
coef	std err	z	P> z		[0.025	0.975]
const	0.0032	0.002	1.364	0.172	-0.001	0.008
SPX_excess	0.5037	0.062	8.168	0.000	0.383	0.625
SENT_ORTH	0.0008	0.005	0.151	0.880	-0.009	0.011
Interaction	0.0002	0.007	0.027	0.979	-0.013	0.014
Omnibus:		2.631	Durbin-Watson:		2.037	
Prob(Omnibus):		0.268	Jarque-Bera (JB):		2.356	
Skew:		-0.157	Prob(JB):		0.308	
Kurtosis:		3.312	Cond. No.		23.6	

Notes: [1] Standard Errors are heteroscedasticity robust (HC3)

5.6 FF5 robustness

After controlling for Fama-French five factors, the traditional risk factors dominate the explanatory power of the regressions. Sentiment coefficients become uniformly insignificant with only a marginal effect for XLE around the 10% level and behavioral beta does not survive once size, value, profitability and investment factors are included. After controlling for Fama-French factors the results imply that sentiment does not provide robust incremental explanatory power beyond standard risk factors, so H6 is rejected.

Table 8.Fama-French five factor plus sentiment regressions.

ETF	Sentiment Coeff	Sentiment pvalue	R squared
0 XLE	0.010887	0.084075	0.426562
1 XLF	0.003369	0.372912	0.711335
2 XLP	0.001133	0.717056	0.468551
3 XLK	-0.004739	0.381138	0.748713
4 XLU	0.001726	0.655186	0.256902

5.7 Economic Significance and Pooled Panel Analysis

The pooled panel regression, which stacks all ETF observations, shows that the market factor is highly significant, whereas the sentiment coefficient is statistically insignificant ($p = 0.17$), with an R^2 of 48%. This suggests that, at the aggregate ETF market level, returns are primarily driven

by market risk and sentiment plays only a minor role, providing no strong evidence of an aggregate priced sentiment effect and leading to rejection of H7 in this dimension.

Table 9. Pooled panel regression of ETF excess returns on market and sentiment.

OLS Regression Results						
Dep. Variable:	excess	R-squared:	0.483			
Model:	OLS	Adj. R-squared:	0.482			
Method:	Least Squares	F-statistic:	363.8			
Date:	Thu, 05 Mar 2026	Prob (F-statistic):	1.42e-128			
Time:	11:46:34	Log-Likelihood:	2514.0			
No. Observations:	1440	AIC:	-5022.			
Df Residuals:	1437	BIC:	-5006.			
	Df Model:		2			
	Covariance Type:		HC3			
coef	std err	z	P> z		[0.025	0.975]
const	0.0018	0.001	1.648	0.099	-0.000	0.004
SPX_excess	0.9217	0.034	26.921	0.000	0.855	0.989
SENT_ORTH	0.0028	0.002	1.371	0.170	-0.001	0.007
Omnibus:	146.063	Durbin-Watson:	2.055			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	880.666			
Skew:	0.242	Prob(JB):	5.83e-192			
Kurtosis:	6.800	Cond. No.	23.2			

Notes: [1] Standard Errors are heteroscedasticity robust (HC3)

5.8 Removing Market beta

The behavioral beta-only regression indicates that investor sentiment does not significantly explain ETF excess returns in the absence of market controls. With the exception of XLK, whose sentiment coefficient is statistically significant but small, sentiment coefficients are insignificant and R^2 values remain very low below 4% for all ETFs. This suggests that investor sentiment cannot independently price sector ETFs and its effect is largely conditional on market risk factors, so H8 is rejected.

Table 10. Behavioral beta only regressions (no market factor).

ETF	Alpha	Behavioral Beta	Sentiment pvalue	R squared
0 XLE	0.007267	0.001451	0.828665	0.000179
1 XLF	0.006001	-0.007267	0.157196	0.006360
2 XLP	0.005830	-0.003608	0.307011	0.004716
3 XLK	0.009588	-0.019003	0.037780	0.038816
4 XLU	0.006044	-0.004260	0.266389	0.004346

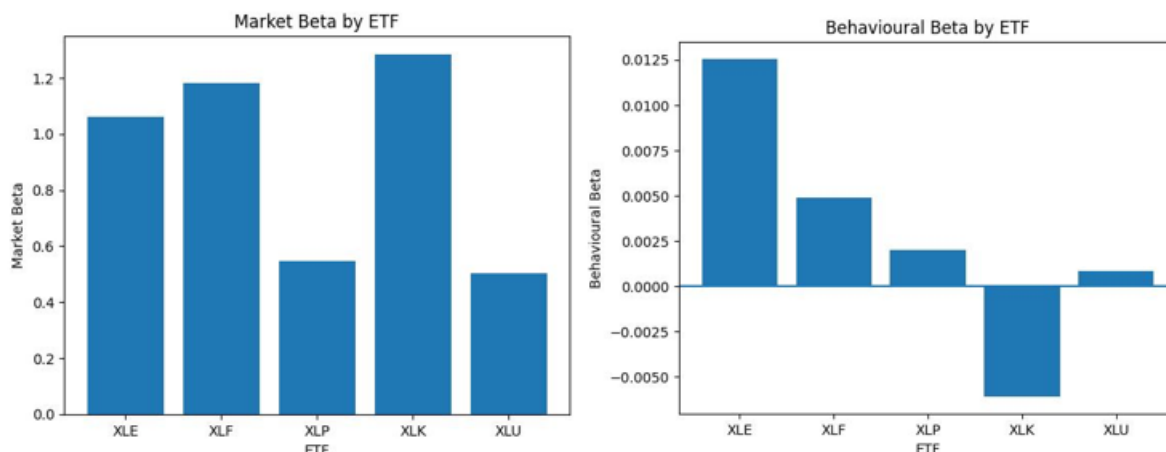


Figure 3. Behavioral beta estimates and Market beta Estimates for Sector ETFs.

6. Discussion

6.1 Key Patterns and Underlying Mechanisms

Taken together, the empirical tests indicate a consistent pattern: market risk remains the dominant driver of sector ETF returns, whereas sentiment-related risk appears weak and is not robustly priced at the sector level. Market betas are strongly significant and provide substantial explanatory power. In contrast, extending the model to include a behavioral component only marginally improves model fit and yields statistical significance in a single sector (energy), where the estimated effect remains economically small. Further evidence reinforces this pattern. Predictive regressions, portfolio spread tests, regime interaction models, multifactor robustness checks, pooled regressions, and behavioral-beta-only specifications were also examined.

Across these approaches, the results consistently point to the same conclusion. At the frequency examined here, the behavioral factor does not produce a stable return premium or a strong forecasting signal for sector ETF excess returns.

Several mechanisms may help explain these findings. First, sector ETFs aggregate large numbers of underlying firms, which means that firm-specific sentiment-driven shocks are diversified at the portfolio level. Only broad market-wide sentiment effects are likely to remain after aggregation, and these are already partially captured by traditional risk factors.

Second, sector ETFs are widely traded by institutional investors. Active arbitrage by these participants can correct mispricing relatively quickly and reduce the persistence of behaviorally driven deviations, particularly in large and liquid U.S. markets.

Third, the monthly sampling frequency may smooth short-lived behavioral influences that arise around news releases or macroeconomic announcements. Any intraday or weekly effects are likely to dissipate before appearing as systematic patterns in monthly excess returns.

6.2 Relation to Behavioral Beta Evidence in Prior Studies

Relative to many stock-level studies documenting behavioral beta effects, the evidence observed at the sector ETF level is noticeably weaker. Prior research often reports pronounced regime dependence and, in some cases, flattened or inverted Security Market Lines during high-sentiment periods. The results presented here, however, show little evidence that such dynamics are strongly reflected in sector ETF returns. This aligns with the notion that mispricing due to behavioral factors is most apparent where arbitrage constraints are tighter. Small, illiquid, or difficult-to-short stocks are typically more susceptible to behavioral distortions than diversified, institutionally dominated sector investment vehicles. Consequently, aggregation within ETFs may substantially reduce the observable impact of behavioral factors.

Another factor relates to the design of many earlier studies. A number of behavioral beta papers examine portfolios sorted on characteristics that are highly sensitive to investor sentiment, or they focus on high-frequency reactions to sentiment shocks. In contrast, the present analysis evaluates broad sector ETFs using monthly data. Differences in both aggregation and frequency naturally attenuate measured behavioral effects.

The finding that the Behavioral component loses statistical significance once Fama-French factors are introduced is also consistent with previous research suggesting that some apparent Behavioral patterns are partly captured by established factor structures. Rather than contradicting Behavioral asset-pricing theory, the results suggest that at the sector ETF level any remaining Behavioral influence is either small, rapidly arbitrated away, or closely related to existing risk factors. More broadly, this highlights an important methodological consideration: evidence for Behavioral beta may depend strongly on the level of aggregation, the composition of the investor base, and the frequency at which returns and Behavioral indicators are measured.

6.3 Economic Implications for Investors and Modelling

From an investment perspective, the results indicate that sentiment-tilted sector allocation strategies provide limited incremental benefit when evaluated at a monthly horizon. Market beta

and conventional risk factors account for the majority of variation in sector ETF returns, whereas the Behavioral component does not generate a reliable risk premium or a strong forecasting advantage, particularly once transaction costs and model uncertainty are considered.

For investors evaluating sector allocation decisions, these findings suggest that greater emphasis may be placed on factor exposures, diversification, and macro-driven risk management rather than systematically overweighting sectors with higher Behavioral sensitivity.

From a modelling standpoint, the results imply that adding a simple linear Behavioral factor to monthly sector ETF regressions does not materially improve asset-pricing performance relative to standard CAPM or Fama-French specifications. Future research attempting to capture Behavioral effects at this level may require richer modelling approaches, such as time-varying betas, nonlinear relationships, or higher-frequency measures of returns and Behavioral indicators.

In practice, Behavioral influences may still be relevant in contexts such as intraday risk management, options markets, or stock-selection strategies. However, when focusing on broad U.S. sector ETFs and monthly excess returns, the evidence presented here suggests that traditional risk factors remain the primary drivers of pricing.

7. Limitations and Areas for Further Research

7.1 Limitations

Although this study provides valuable insights, several limitations should be noted. First, only five sector ETFs are analysed, which might not capture the full cross-sectional variation of the broader equity market. Second, the use of monthly data could attenuate short-term Behavioral effects that might be more evident at higher frequencies. Third, the reliance on a single orthogonalized sentiment index could introduce measurement error and might not capture all facets of investor sentiment. Fourth, the study focuses exclusively on the U.S. equity market, limiting the generalisability of the findings to other countries or market structures. Finally, the absence of intraday or high-frequency data prevents an examination of rapid sentiment-driven trading dynamics and short-term price adjustments around news or macroeconomic events.

7.2 Areas for Further Research

Future studies could extend the analysis by examining individual stocks or more narrowly defined ETFs to better capture cross-sectional variation in Behavioral exposure. Incorporating alternative

sentiment proxies, such as survey-based measures, options-implied indicators, or text-based sentiment from news and social media could provide a more complete understanding of Behavioral influences. Employing higher-frequency return and sentiment data might identify short-term sentiment effects that are not observable at the monthly level and are quickly arbitrated away. Additionally, extending the analysis to international markets would enable cross-country comparisons and may clarify whether sentiment effects vary across institutional settings, cultural contexts, and regulatory regimes.

8. Conclusion

This study investigates whether investor sentiment and Behavioral beta are systematically priced in U.S. equity sector ETFs and whether Behavioral beta provides incremental information beyond traditional market beta. Using approximately twenty-three years of monthly data for five major sector ETFs, the study conducts a series of empirical tests, including CAPM, Behavioral beta regressions, predictive regressions, high-low Behavioral-beta spread tests, regime interaction models, Fama-French five-factor robustness checks, pooled panel regressions, and Behavioral-beta-only specifications. The results strongly support the standard CAPM benchmark (H1), with market beta significantly explaining sector ETF returns. In contrast, there is only weak and sector-specific evidence that Behavioral beta is priced (H2), and minimal support for predictive, spread, regime-dependent, multifactor-robust, pooled, or market-independent sentiment effects (H3-H8).

Controlling for Fama-French five factors and pooled panels renders sentiment coefficients statistically insignificant, and Behavioral beta does not survive as an independent priced factor. This indicates that exposure to market risk does not automatically translate to exposure to sentiment risk. Overall, the findings suggest that sentiment-related risk is not a robustly priced dimension at the sector ETF level, and traditional risk factors remain the dominant drivers of monthly sector ETF returns in the U.S. market. For investors, sentiment-tilted sector allocation strategies offer limited incremental benefit over standard factor-based approaches at this horizon, particularly in a market segment with high liquidity and substantial institutional participation. Future research could extend this framework to individual stocks, alternative sentiment measures, higher-frequency data, and international markets to better identify circumstances under which sentiment risk becomes economically significant.

References

1. Antoniou, C., Doukas, J., & Subrahmanyam, A. (2016). Investor sentiment and the cross-section of stock returns. *Journal of Financial Economics*, 119(2), 251-272. <https://doi.org/10.1016/j.jfineco.2015.12.005>
2. Baker, M., & Wurgler, J. (2006). Investor sentiment and the cross-section of stock returns. *Journal of Finance*, 61(4), 1645-1680. <https://doi.org/10.1111/j.1540-6261.2006.00885.x>
3. Baker, M., Wurgler, J., & Yuan, Y. (2012). Global, local, and contagious investor sentiment. *Journal of Financial Economics*, 104(2), 272-287. <https://doi.org/10.1016/j.jfineco.2011.05.008>
4. De Long, J. B., Shleifer, A., Summers, L. H., & Waldmann, R. J. (1990). Noise trader risk in financial markets. *Journal of Political Economy*, 98(4), 703-738.
5. Fama, E. F., & French, K. R. (1992). The cross-section of expected stock returns. *Journal of Finance*, 47(2), 427-465. <https://doi.org/10.1111/j.1540-6261.1992.tb04398.x>
6. Frijns, B., Huynh, T. D., Tourani-Rada, A., & Westerholm, P. J. (2018). Institutional trading and asset pricing. *Journal of Banking & Finance*, 87, 90-105. <https://doi.org/10.1016/j.jbankfin.2017.10.008>
7. Glushkov, D. (2018). Sentiment beta: Measuring the sensitivity of returns to investor sentiment. *Journal of Behavioral Finance*, 19(3), 234-249. <https://doi.org/10.1080/15427560.2018.1465902>
8. Kenneth R. French Data Library. (n.d.). Fama/French 5 factors [Data set]. Dartmouth College. Retrieved March 2026, from http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

9. Shen, J., Yu, J., & Zhao, S. (2017). Sentiment regimes and macro risk premia. *Journal of Behavioral Finance*, 18(3), 247-257. <https://doi.org/10.1080/15427560.2017.1285412>
10. Verma, R., & Soydemir, G. (2009). The impact of individual and institutional investor sentiment on the market price of risk. *Journal of Economics and Finance*, 33(2), 112-129. <https://doi.org/10.1007/s12197-009-9050-3>
11. Wang, W. (2019). Institutional investor sentiment, beta, and stock returns. *International Review of Financial Analysis*, 63, 101-115. <https://doi.org/10.1016/j.irfa.2018.11.004>
12. Wang, W., & Duxbury, D. (2021). Institutional investor sentiment and the mean-variance relationship: Global evidence. *Journal of Empirical Finance*, 62, 1-17. <https://doi.org/10.1016/j.jempfin.2021.07.004>
13. Yu, J., & Yuan, Y. (2011). Investor sentiment and the mean-variance relation. *Review of Financial Studies*, 24(8), 2443-2477. <https://doi.org/10.1093/rfs/hhr010>